

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of B Sc Biotechnology Examination, September, 2019

Course code & Name: BT406: Bioprocess Technology

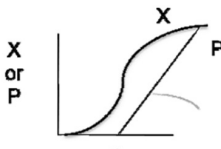
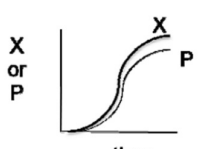
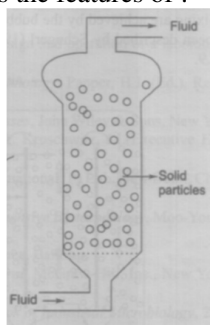
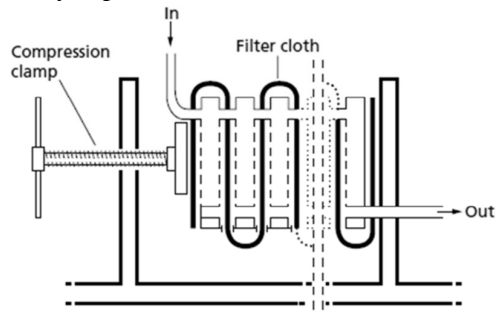
Date: 23.09.2019 Day: Monday Time: 10.00 a.m to 10.30 a.m Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	Molasses is byproduct of which of the following industry? (i) sugar (ii) paper (iii) starch (iv) dairy	
2.	In bioreactors, the main role of an impeller is to provide: (i) foam control (ii) mixing (iii) temperature control (iv) none of the above	
3.	Which of the following is a physical mutagen used for strain improvement? (i) UV (ii) x-rays (iii) gamma rays (iv) all of the above	
4.	“substances added prior to or simultaneously with the fermentation which are incorporated without any major change into the molecule of the fermentation product and which generally serve to increase the yield or improve the quality of the product” denotes: (i) inducers (ii) precursors (iii) vitamins (iv) inhibitors	
5.	In bioreactors, the role of baffles is to help in: (i) foam control (ii) mixing (iii) aeration (iv) none of the above	
6.	The term rheology denotes the study of: (i) fluid flow behavior (ii) mixing (iii) osmotic pressure (iv) heat transfer	
7.	The following statement is NOT TRUE for heat transfer in bioreactors (i) Heat transfer $\propto \Delta T$ (temperature difference) (ii) Heat transfer $\propto A$ (contact area) (iii) heat transfer depends on both ΔT and A (iv) heat transfer depends on Bioreactor type	
8.	The purpose of immobilization of enzymes is to: (i) reuse the enzyme (ii) increase enzyme density in a location (iii) improve the stability of the enzyme (iv) all of the above	
9.	The following rheological behavior is NOT observed in fermentation broths: (i) Newtonian (ii) pseudoplastic (iii) Bingham plastic (iv) dilatant	

10.	The immobilization of cells and enzymes using sodium alginate follows which of the following mechanisms? (i) adsorption (ii) encapsulation (iii) entrapment (iv) covalent coupling	
11.	One of the following is a serious problem in SSF: (i) Low capital investment (ii) Low water utilization (iii) Absence of foam (iv) heat and mass transfer	
12.	Which of the following is an <i>external sensor</i> in a bioreactor which are present in or above the fermentation medium? (i) oxygen probe (ii) pH probe (iii) temperature probe (iv) impeller speed meter	
13.	Which figure among the figures given below best describes the non-growth associated product formation kinetics? (X=biomass, P=Product)  (i)  (ii)	
14.	Which of the following is a high price microbial and animal products? (i) insulin (ii) ethanol (iii) citric acid (iv) mushroom	
15.	Which among the following products is a product obtained from plant cell culture? (i) monoclonal antibodies (ii) Taxol (iii) penicillin (iv) none of the above	
16.	Which among the following products is a product obtained from animal cell culture? (i) monoclonal antibodies (ii) Taxol (iii) penicillin (iv) none of the above	
17.	The following figure correctly depicts the features of : 	
18.	The following is/are a major advantage of SSF: (i) Low capital investment (ii) Low water utilization (iii) Absence of foam (iv) all of the above	
19.	The following figure correctly depicts: 	
20.	Which of the following method(s) is a mechanical method used for cell disintegration? (i) ultrasonication (ii) French Press (iii) Bead mills (iv) all of the above	

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Date: 23.09.2019 Day: Monday Time: 10.30 a.m to 01.00 p.m Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed		20
Q-II (A)	Answer any one of the following:	04
	(i) What is primary and secondary screening? Giving an example, describe in a step-wise manner screening of microorganisms producing an industrially important product.	
	(ii) What is the purpose of heat transfer in bioprocess technology. Describe jacketed kettle as heat transfer method in bioreactors.	
Q-II (B)	Answer any two of the following:	06
	(i) Giving diagram explain difference between Newtonian and non-Newtonian fluids.	
	(ii) Describe any one method for continuous sterilization of fermentation medium	
	(iii) Discuss the role of cost of fermentation media in the overall economics of a fermentation process.	
Q-II (C)	Answer any one of the following:	04
	(i) Briefly discuss the role of algae and fungi in bioprocesses.	
	(ii) What is the need for improvement of industrial strains? Discuss any one method for improvement of industrial strains.	
Q-II (D)	Describe any two of the following:	06
	(i) Filters used for cell separation in bioprocess industries.	
	(ii) Chromatographic method for product recovery	
	(iii) Mechanical methods for cell disruption.	

SECTION - II		
Q-III	Answer the following questions as directed	30
QIII-(A)	Answer any one of the following: (i) Briefly discuss any two cheaper raw material used for formulation of industrial scale production media. (ii) Compare media used for plant and animal cell cultures.	04
Q-III (B)	Answer any one of the following: (i) What are the characteristics of bioprocesses as compared to chemical processes? (ii) Explain briefly types of sensors used in control of bioprocesses.	04
Q-III (C)	Answer any two of the following: (i) Giving suitable diagrams describe air-lift reactors. (ii) Enlist methods for immobilization of cells. Briefly explain any one method immobilization of cells. (iii) Describe the method for inoculum development for fungal fermentations.	06
Q-III (D)	Answer any two of the following: (i) Give a labelled diagram for a fermenter used for aerobic cultivation of microorganisms. What is the role of antifoams? (ii) Giving an example discuss the use of solid substrate fermentation. (iii) What are spargers? Give labelled diagram for any three types of spargers.	06
Q-III (E)	Answer any one of the following: (i) Describe bacterial growth curve. (ii) Discuss kinetics of growth associated product formation with suitable examples.	04
Q-III (F)	Write a note on any two of the following: (i) Plant cell culture (ii) Cultivation of microalgae (iii) Animal cell cultivation	06